STRUCTURE AND PHYSIOLOGY OF ECTOMYCORRHIZAE

I. THE PROCESS OF MYCORRHIZA FORMATION IN NORWAY SPRUCE *IN VITRO*

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SUMMARY

The process of mycorrhiza formation in previously uninfected roots was studied *in vitro*, using the fungus *Piloderma croceum* Erikss. & Hjortst. and the Norway spruce, *Picea abies* (L.) Karst. as model organisms. The process could be divided into the following phases: fungal growth stimulation by root metabolites; formation of a hyphal envelope on the root; intercellular penetration by single hyphae, change in fungal morphology into labyrinthic tissue formation leading to Hartig net formation, and extension of labyrinthic tissue to form a mantle. *P. croceum*, growing in dead tissue, showed saprophytic growth *in vitro*, a capacity suppressed in living tissue. The way of fungal penetration appeared to be mechanical. A hypothetical model for host-fungus interactions regulating the mycorrhiza infection process is also discussed.

INTRODUCTION

The technique of ectomycorrhiza synthesis *in vitro* has been known since Melin's (1923) pioneer work in the 1920s. Many studies of the anatomy of ectomycorrhiza have also been published since then (review by Marks and Foster, 1973). However, the process of mycorrhiza formation has never been studied in detail *in vitro*, and most of the information from field studies is based on circumstantial evidence only (Marks and Foster, 1973; Slankis, 1973), or concerns the extension of existing mycorrhizal infection rather than *de novo* formation (Robertson, 1957).

The fungus, *Piloderma croceum* Erikss. & Hjortst., (= *Corticium bicolor* Peck) and its mycorrhizal host, the Norway spruce, *Picea abies* (L.) Karst., have been employed as a model system for studies on ectomycorrhizae, the organisms being identifiable in the field, and cultivable *in vitro*. In a previous paper (Nylund, 1980), structural aspects of natural and synthesized mycorrhizae with the two organisms are described. In the present study, the details of the process of mycorrhiza formation in previously uninfected roots were studied *in vitro*, using a partly new procedure for the synthesis.

MATERIALS AND METHODS

Spruce seedling material was produced with hypochlorite sterilized seed, all from the same lot, germinated on 2% malt extract agar. *P. croceum* was isolated from yellow spruce mycorrhizae, and kept in stock on 2% malt extract agar. The fungus

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is very stable in culture, Mikola's (1967) isolate still retaining its virulence after 14 years. The identity of the isolate was established by comparing, in a spectrophotometer, ethanol extracts of fungal pigment with extracts of strains from Mikola and Zak (1976), with reference to Erdtman's (1948) analysis of the yellow pigment. Isolates of other fungi used in the study were taken from the Department's culture collection.

Radicles of germinating seed and very young seedlings were studied on agar dishes with either Melin's medium as modified by Krupa and Fries (1971) but with ammonium tartrate as a source of nitrogen, or with Ingestad's (Andersson *et al.*, 1974; Ingestad, 1970, 1971) mineral medium A+B (designed for tree seedling culture with continuous nutrient supply). The plates were kept standing on their edge in stacks, at uncontrolled room light and 20 °C.

Seedlings aged 2 to 4 months were studied in specially designed 28 × 250 mm test tubes with a 19 × 30 mm tube through the side wall, about 150 mm from the bottom (Fig. 1). Two-thirds of the tube periphery opposite the side tube was

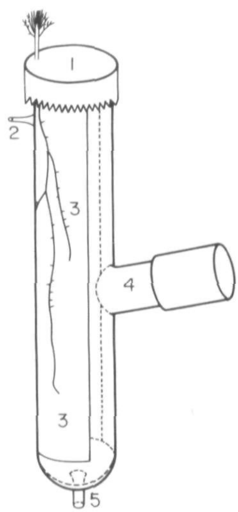

Fig. 1. The culture vessel. 1, Aluminum foil cap; 2, inlet from pump; 3, filter paper pad; 4, capped side tube; 5, outlet. (2 and 5 only in tubes for continuous nutrient supply).

covered with a chromatography paper pad, reaching from top to bottom. The tube was capped with cotton and aluminium foil. Upon transplanting, the radicle of a sterile seedling was inserted through a small hole in the foil either on the inner surface of the pad or between the wick and the glass, and the passage was sealed with silicon grease. Melin's modified medium (40 ml) was added through the side tube, provided with a test tube cap, and the solution was changed monthly. When the seedling was established and the radicle had reached a length of about 50 mm, the tube was inoculated with stock culture mycelium (3 mm mycelium agar plug) through the side tube.

The seedlings were grown in climate chambers at 20 °C, 60% r.h. and continuous light (2200 lux, 79·2 μE m^{-2} s^{-1}) from Osram L Fluora tubes. Only the shoot parts were illuminated, the roots were shaded from direct light. Theodorou and Bowen (1971) found that mycorrhiza formation was maximal at that temperature. Harley (1969) showed 20°C to be optimal for the growth of a number of mycorrhiza-forming plants. Spruce grows well at that temperature, and with a long day or continuous light regime never forms terminal buds or stops growing (unpublished results).

To improve shoot growth, which was poor compared to normal glasshouse seedlings, the system was modified to make possible continuous nutrient supply. The tubes were fitted with capillary inlets at the top and outlets in the bottom (Fig. 1), and were irrigated (about 10 ml) with Ingestad's medium once daily. Shoot growth improved markedly, but severe contamination problems limited the application of the technique in the present study.

For transmission electron microscopy (TEM), root material was processed in glutaraldehyde, osmium tetroxide and Epon, with lead and uranyl post-staining. It was very difficult to obtain good embedding results.

For light microscopy (LM), Epon sections were used. However, glycol methacrylate with 5% polyethylene glycol 400 and 0·4% azobisbutyronitrile (Bennett *et al.*, 1976) proved to be a superior medium for embedding and sectioning. Further details on microscopy techniques will be discussed in a subsequent paper (Nylund, in prep.).

Results and Discussion

Fungal growth stimulation by roots

Near the roots of spruce seedlings, the growth of *P. croceum* was notably stimulated. Germinating seed and 2 to 4 cm radicles on Melin and malt agar dishes, both containing carbohydrates, stimulated the radial growth of the fungus. Since the seedlings grew at a light intensity below the compensation point, and the media contained sufficient carbohydrate for good fungal growth, the effect must be due to factors other than supply of photosynthetic carbohydrates. The stimulation was discernible only on few millilitres away from the root. Once the fungus reached the root, it formed a hyphal envelope of denser growth over it, and it also grew faster along the radicle than over the agar surface.

The roots of tube seedlings (Melan's medium) exerted a similar influence on *P. croceum*. When the fungus was inoculated near a root, it grew excentrically on the pad, with rates two or three times faster along the root than away from it. When roots and inocula were located on different sides of the paper wick, the fungus formed a concentric colony on the inoculum side, but penetrated the paper and appeared on the opposite side only adjacent to the roots.

The rate of stimulation was highly variable in both cases. Sometimes, where the root was poorly developed, no stimulation could be observed. Older parts of radicles where the cortex was dead stimulated the extension growth of the fungus but no hyphal envelope was produced. On the distal parts of growing roots, the formation of a hyphal envelope was the rule (Fig. 2), and always preceded penetration of the host tissue.

On dishes (low light) with the carbohydrate-free Ingestad's medium, the growth of the fungus was very poor. Once the sparse and colourless hyphae reached the root, however, they formed a thin envelope and soon turned yellow.

The inoculations of tube seedling cultures (full light) on Ingestad's mineral medium produced, after a very long incubation time (2 months), well developed hyphal envelopes. In this case, the light supplied was adequate. The spread of the fungus was very slow, however; the amount of photosynthetic carbohydrate released by the seedling was apparently insufficient to support much fungal growth on the pad beyond the area in contact with the root.

Parallel radicle and tube cultures inoculations with *Suillus bovinus*, another

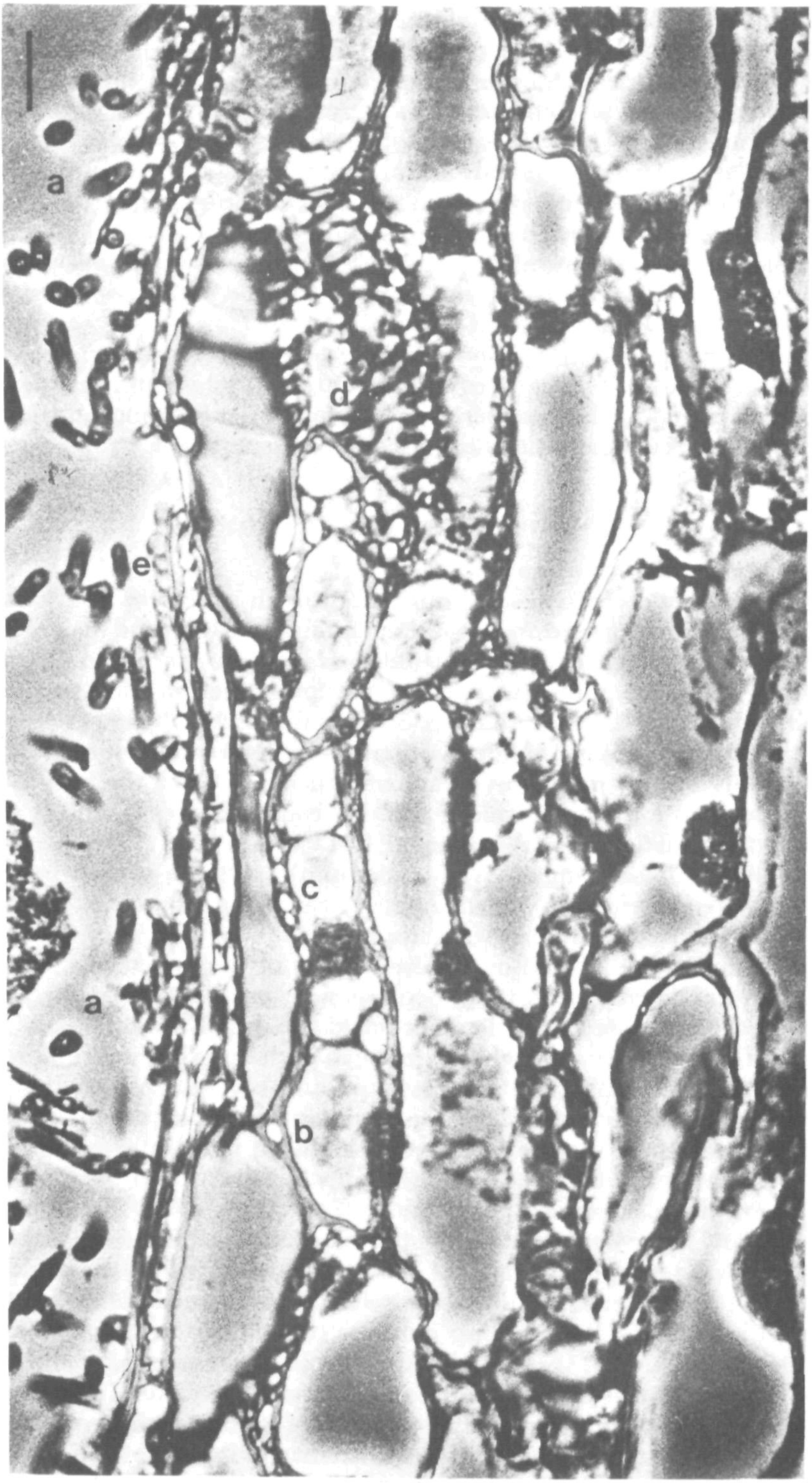

Fig. 2

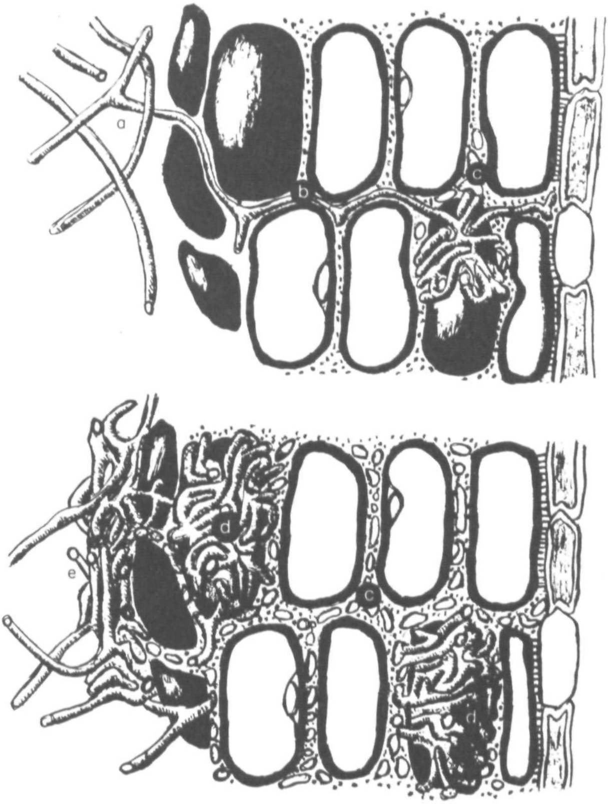

Fig. 3. Diagrammatical presentation of the infection process. The lettering corresponds to Figure 2. (a) Single hyphae form an envelope around the root, (b) penetrate middle lamellae in the cortex, (c) and start branching to form ladder-like structures as seen in face view, (d) the branching leads to the formation of a new type of fungal tissue, characterized as 'labyrinthic', which makes up the Hartig net, and ultimately, (e) the mantle. See also Table 1 and Figure 4.

mycorrhiza former (Melin's medium), revealed no distinct stimulation of fungal growth, but a hyphal envelope was formed on the root. It is possible that *S. bovinus*, being a much faster grower in axenic agar culture than *P. croceum*, was less dependant on some growth factor than the latter fungus.

None of the common soil parasites and saprophytes, *Heterobasidium annosum*, *Trichoderma viride*, *Mycelium radicis atrovirens*, three *Penicillium* spp. and a *Mortierella* sp. showed any sign of growth stimulation on or around the roots in corresponding experiments with radicles on agar plates.

Bowen and Theodorou (1973 and previous work) found stimulatory effects on mycorrhizal fungi near and on roots in soil and related it to Melin's M-factor. The

Fig. 2. Light micrograph (phase contrast) of a spruce short root inoculated with *Piloderma croceum*, longitudinal section. This micrograph shows the stages of infection diagrammatically illustrated in Figure 3. (a) The root is covered with an envelope of free hyphae, not closely attached to the surface or to each other. (b) Single hyphae penetrate into the middle lamellar matrix. After some time they start branching profusely, to producing a ladder-like effect in cross section (c), and a labyrinth pattern in face view (d). Finally, (e) labyrinthic tissue forms the mantle, which in cross section can be distinguished from the hyphal envelope by showing different light refracting and staining characters, and by the cells being closely attached to the surface and to each other (for face view, see [Fig. 5(b)–(f)]. Note that the cortical cells are nucleated, and hence living. Scale = 10 μm.

effect observed in the present study is also believed to be, at least in part, identical with Melin's (1963) M-factor. In the same way exactly as in our study, the M-factor stimulated moderate to slow growers more than fast ones, and it was not an effect of carbohydrate nutrition only. Melin's most interesting point was that even tomato roots produced substantial amounts of M-factor. The suggestion that it may be a common root metabolite, contrasts strongly with the specific response by mycorrhizal fungi only, the non-mycorrhizal ones being unaffected. Lindeberg *et al.* (1980) recently demonstrated similar selective effects in litter, where minor constituents of the plant material (taxifolin glucoside, etc.) selectively stimulated fungal decomposers of that particular kind of litter. In the present case, however, mycorrhizal fungi have adapted to react to a factor or factors from both host and other plant roots, indicating the existence of further selective factors.

In the present study, no attempt was made to investigate further this stimulation of growth, through which the first essential phase of mycorrhiza formation was brought about: the formation of a hyphal envelope on the root.

Fungal penetration of short roots

For the study of the penetration process, newly formed short roots, 2 mm long, of fully illuminated plants in tubes on Melin's liquid medium inoculated with *P. croceum*, were employed as a standard. Figure 2, a longitudinal section through such a root inoculated with *P. croceum*, provides a comprehensive view of the processes discussed in the following; Fig. 3 is a diagrammatic interpretation of that plate.

The formation of a hyphal envelope on the root was a prerequisite of fungal penetration of the host. An envelope of this kind developed with *P. croceum* within 2 to 3 weeks after the first contact between root and fungus. However, the time required varied considerably, depending on the condition of the host; it should be noted that the hyphal envelope discussed here was not identical with the normal mantle; it always consisted of free, sparsely branched hyphae, which were never closely attached to the root surface.

Single hyphae, originating from this envelope, then penetrated into the root at several points. In the present experiments this took place all over the short root except at the apical meristem and in the differentiation zone (Fig. 2). All the root cortex was nucleated and living (Fig. 2). At the time of penetration, no morphological changes in the host tissue or cells could be observed, as compared with uninfected parts of the root or uninoculated roots.

Penetration took place only in the mature cortex and without the formation of any kind of appressorium or comparable structure (Fig. 2). The hyphae simply pushed their way into the cortical middle lamella, after having penetrated between the dead root cap cells on the surface. Penetrating hyphae were found both in the middle lamellae between two adjacent cells and in the intracellular spaces, being filled with pectic material in the short root cortex (Fig. 2). The whole process appeared to be predominantly mechanical in character: the hyphae, almost angular or wedge-shaped in cross section, appeared to expand in the host tissue, thus separating the host cells, and creating space for further expansion [Fig. 4(a) and (b)]. The intercellular middle lamella, even immediately adjacent to expanding hyphae but outside the splitting zone, was also entirely unchanged in structure, as was the cortex cell wall fine structure [Fig. 4(a)].

Foster and Marks (1966, 1967) were the first to propose a substantiated theory concerning the mode of fungal penetration during mycorrhiza formation. They

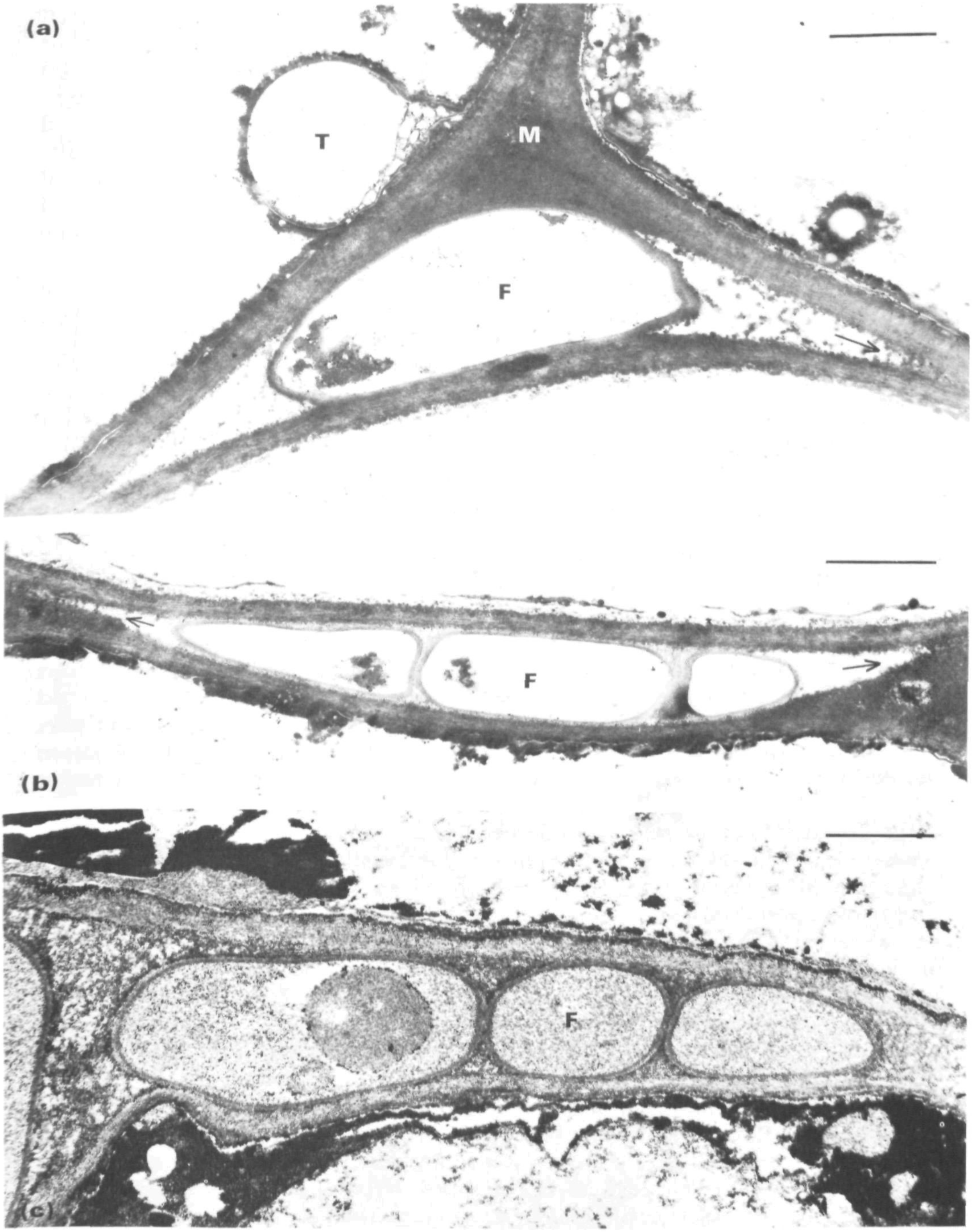

Fig. 4. Transmission electron micrographs of the same material. (a) A single hypha (F) penetrating between three adjacent cortical cells. Note that the middle lamellae matrix (M) in the intercellular space is in no way degraded or visibly affected by the fungus, which seems to split apart the cells by mechanical force. T, An artifact, consisting of tannin on the tonoplast. (b) The hypha has branched into three; the side cells are wedge-shaped and will, upon expansion, further split the middle lamella (arrows). Neither in (a) or (b) do the cell walls seem to be disorganized or damaged. (c) The expansion of the fungal cells is about to cease, and the free intercellular space is being filled up with flocculent material, which later will turn into a compact matrix. Scale = 1 μm.

concluded on the basis of TEM sections of natural mycorrhizae that the penetration process is mainly of mechanical character, and they advanced a hypothesis that osmotic swelling by means of glycogen breakdown may be the driving force. The observations made in the present study are entirely in line with these findings.

After having penetrated two to three cell layers deep, the single hyphae started branching profusely in the tissue, without 'regular' formation of septa. The result, seen in 'front view' [Fig. 5(a)], resembled a maze or labyrinth, with no hyphal structure discernible: instead, a fungal tissue of densely packed irregularly shaped cells was formed. In cross section, however, the structure appeared ladder-like [Fig. 4(b)], and could easily be misinterpreted as a series of hyphal coils around a cortical cell. At the margins of the branching structures, the fungal cells were still wedge-shaped, and appeared to split up middle lamellae by swelling [Fig. 4(b)], creating space for further branching (cf. Marks and Foster, 1973). Once the expansion was completed, the fungal cells were rounded off, and the empty intercellular space became filled up with a flocculent material [Fig. 4(c)] which later condensed to form a dense matrix (Nylund, unpublished). From the sites of invasion, the fungus grew in its labyrinthic form in all directions until halted by the endoderm or the undifferentiated tissue near the root apex. The Hartig net of specialized fungal tissue, enclosing all the cortical cells but leaving the plasmodesmic connections intact (Nylund, 1980) was thus formed in response to the close contact with the host.

The labyrinthic form of growth was extended to the surface of the root, and the root was covered with first one and later several layers of labyrinthic tissue, forming a true mantle [Fig. 5(b)–(f)] similar to that observed in natural mycorrhizae. This took place first at patches external to ample Hartig net development (Fig. 2), but later the whole root surface was enveloped by the mantle, including the uninfected apex. At the same time the hyphal envelope grew denser, forming the outer part of the fungal cover, with its projecting hyphal strands, characteristic of *P. croceum* mycorrhizae. Thus, mantle formation followed the development of the Hartig net under our experimental conditions; this also occurred in Petri dish cultures and with carbohydrate-free media. In addition, parallel studies with *P. croceum* on Scots pine, and with *S. bovinus* on spruce, yielded identical results.

In another series of experiments to be reported elsewhere (Nylund *et al.*, in prep.), short roots of young pine and spruce seedlings were inoculated with *P. croceum* and *Pisolithus tinctorius*. In this case, the cortex layer of the roots was very weakened, or even dead. Here, the fungi displayed intracellular growth in dead cells. Furthermore, in living but senescent cells (judging from cytoplasm ultrastructure) typical cell wall papillae (Cf. Aist, 1977) were produced by the host as a response to fungal lysis of the host cell wall. The bearings of these observations will be discussed in next section.

In the present study, where all cortical cells were living, no necrotic or 'hypersensitive' response by the host tissue was observed. On the contrary the lack of visible reaction by the host was striking; the penetrating fungus was just simply 'tolerated'. The cortical hypertrophy (Marks and Foster, 1973) was observed only after the completion of the differentiation of the mantle. In our case the hypertrophy was only a lack of the cortical collapse, always found in uninfected short roots. The non-mycorrhizal fungi discussed above never infected short roots in parallel experiments. Marks and Foster (1973) never recognized the change in fungal morphology, which seems to occupy such a crucial position in the mycorrhiza formation process. Only scattered observations on such fungal

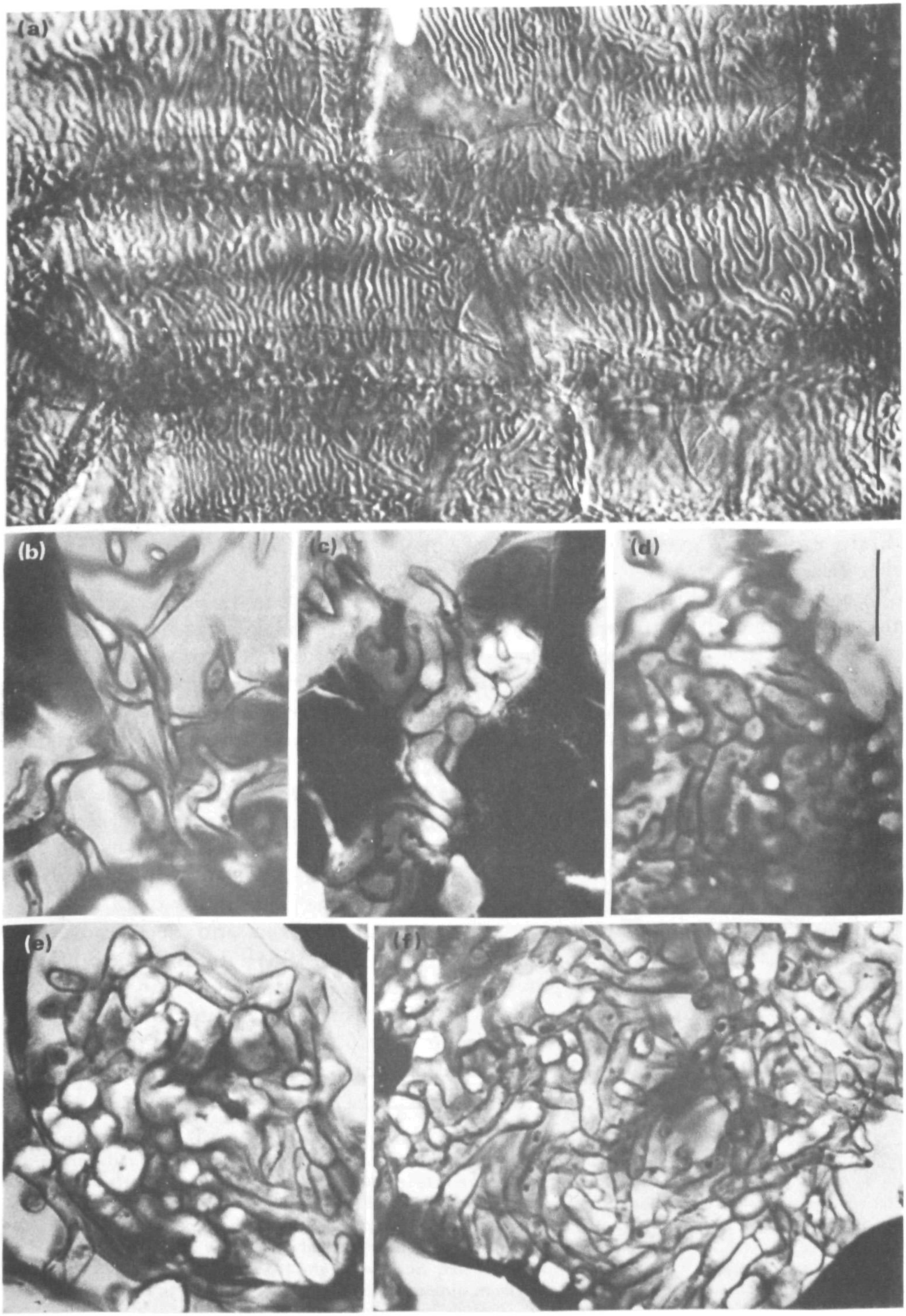

Fig. 5. Light micrographs of Hartig net and mantle in face view. (a) Hartig net in spruce roots, inoculated with *Piloderma croceum*, still lacking mantle. Interference contrast micrograph of cleared smear preparation. The labyrinthic structure of the fungal tissue is evident. (b)–(d) Mantle formation in similar material. The gradual transition from hyphal mode of growth to labyrinthic growth can be seen. Phase contrast, all micrographs from the same short root. Scale = 10 μm.

structures have been made before (Wilcox, 1971; Strullu, 1976) but little attention has been paid to them and no interpretations have been presented.

This morphological change is remarkable since it takes place only after a period of close contact between host and mycobiont, thus having the character of physiologically induced response. The fungus does not change its mode of growth immediately upon coming into close contact with the root surface or even directly after penetration; consequently a prefabricated host metabolite is less likely to be responsible for the change. Also the fungal morphology does not change, even when the fungus becomes subject to pressure in the root cortex, but the hyphal mode of growth continues through several cell layers. It can be speculated that slow mutual interactions of metabolic changes takes place with result in the morphogenesis of the fungus. The metabolic reaction then spreads and finally causes labyrinthic growth even on the surface of the root; thus also inducing mantle formation. However, at some distance from the root surface, varying with the species of fungus, the metabolic influence seems to cease, the hyphal mode of growth is resumed, and a loose outer mantle is formed. (A loose outer mantle is lacking in many types of associations, e.g., *Russula*, unpubl).

Fungal infections of radicles

In radicles of germinating seeds of spruce, 2 to 4 weeks old and inoculated with *P. croceum* and *S. bovinus*, the formation of a hyphal envelope was soon followed by cortical penetration. In radicles, the cortical intercellulars were open and not filled with pectic material, and provided free access for the mycobiont. In this case, the host appeared less able to control the fungus, which rapidly developed a multiseriate Hartig net, breaking many of the plasmodesmic connections between cortical cells (Fig. 6). The spread of the fungus was again halted at the endoderm, but in some cases the fungus penetrated into the apical meristem and soon killed it. No mantle formation was observed.

In parallel inoculations with *H. annosum*, *T. viride*, *M. radicis atrovirens* and the three *Penicillium* spp., none of the fungi entered the cortex, in spite of the open intercellular spaces.

The development of *P. croceum* was also followed on 2 to 4-month-old radicles of tube-grown seedlings. In their proximal parts the cortex was dead although (because of the axenic culture technique) not colonized and decomposed by saprophytes, as invariably found in nature (Robertson, 1954). The hyphal envelope of these parts never became dense and no Hartig net developed in the cortex. Instead, *P. croceum* displayed a saprophytic or nectrotrophic behaviour, penetrating walls and colonizing lumina of the dead cells [Fig. 6(c)]. Even root hair penetration from the outside was observed several times [Fig. 6(d)]. As seen in TEM, the

Fig. 6. (a) The radial growth of *P. croceum* colonies on malt agar in 9 cm Petri dishes is stimulated (arrows) by a young spruce seedling (left) and a germinating seed (right). S, Seed; R, radicle. (b) Transmission electron micrograph of a spruce radicle inoculated with *P. croceum*. The Hartig net is strongly developed, the cortical cells are vital. F. Fungal cells; C, cortical (host) cell. Scale = 10 μm. (c) Light micrograph (phase contrast). *P. croceum* infection of dead cortical cells in a spruce radicle. The fungus completely fills up the outer cortical cells (P) while the inner ones (C) are becoming penetrated (arrow). E, endoderm. Scale = 10 μm. (d) Detail of a root hair (RH) of a dead radicle cortical cell, penetrated from outside by a *P. croceum* hypha (F). The host cell wall is neatly digested at the point of penetration, no pronounced constriction of the hypha can be seen, nor are any fragments of broken wall evident. The tannic deposits in the root hair have folded up (arrows), roughly corresponding to the length of the disappeared cell wall. O, oil droplet. Scale = 1 μm.

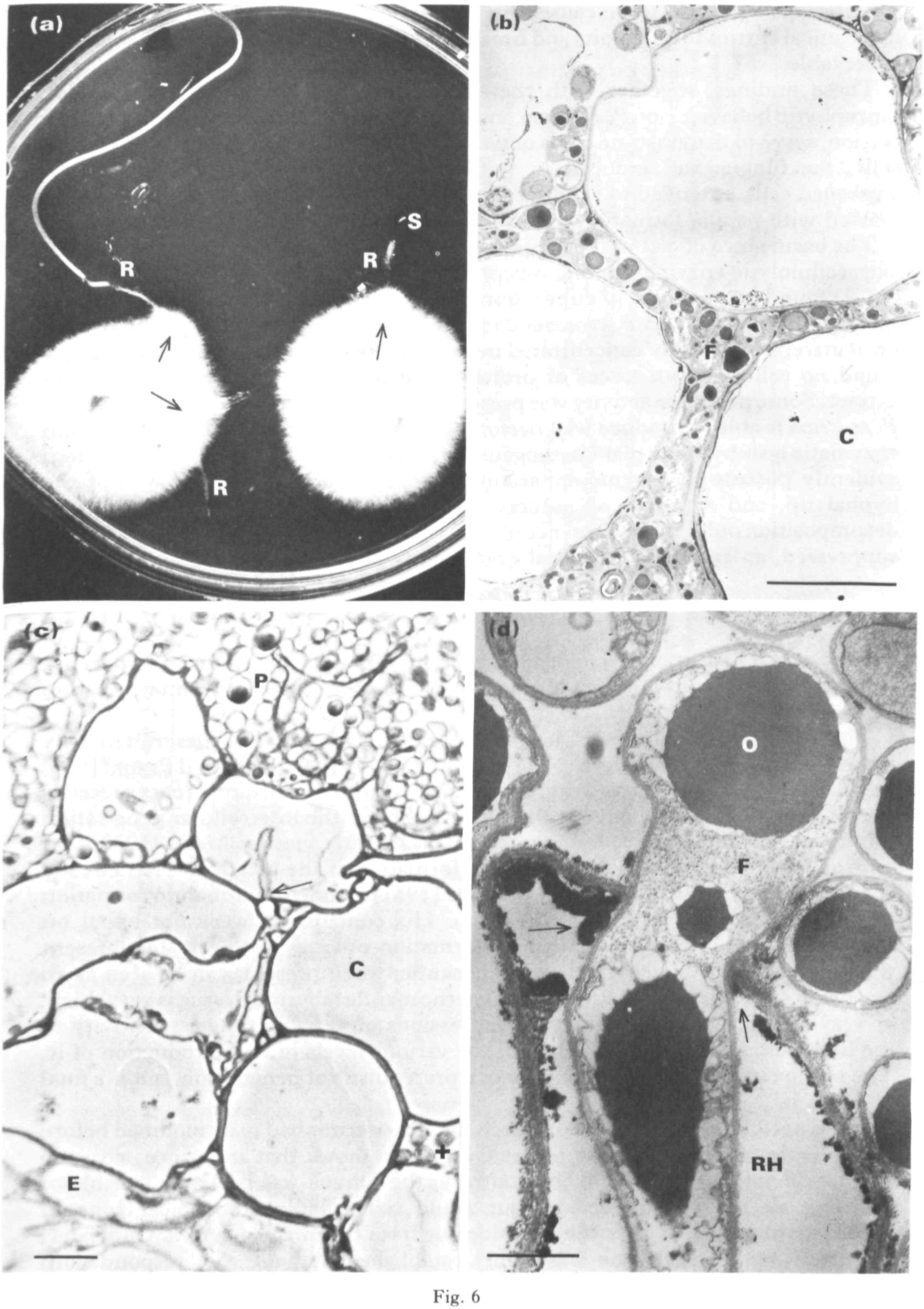

Fig. 6

penetrations appeared to be caused by strictly local cell wall lysis, and not only mechanical entries through pits and breaks. No trace of morphogenetic change was observable.

These findings, together with the observations of weakly parasitic and/or saprophytic behaviour of *P. croceum* and *P. tinctorius* mentioned in the preceding section, serve to demonstrate steps of weakened host control of the fungus: in vital cells, the fungus was induced to 'refrain' from any parasitic behaviour; in weakened cells, it tended to produce intracellular penetrations, to which the host reacted with papilla formation in dead cells, it behaved as a saprophyte.

The occurrence of cell wall perforations raises the question of the occurrence of extracellulolytic enzymes among mycorrhiza formers. Lindeberg and Lindeberg (1977) found no pectinase in cultured mycorrhizal fungi. Ramstedt and Söderhäll (pers. comm.) examined *P. croceum* and *B. variegatus* for cellulase, pectinase and proteinase, using highly concentrated mycelial extracts and culture filtrates. They found no cellulase, but traces of proteinase at least in *B. variegatus* mycelium extract. Some pectinase activity was present in the extracts of both fungi. However, *P. croceum* is able to produce wall perforations, similar to those produced through enzymatic lysis by many plant pathogens (Bracker and Littlefield, 1973). The fungi evidently possess an enzyme apparatus, possibly bound to the cell wall of the hyphal tip, and activated or induced only in dead cortex cells for local wall decomposition only. In the presence of vital cortical cells this capacity is apparently suppressed, at least in the material examined by us.

CONCLUSIONS

The principal phases of mycorrhiza formation in previously uninfected roots are summarized in Table 1, which also provides a slightly speculative interpretation of the process.

The sequence of phases is in general agreement with the descriptions by Robertson (1954) and by Morrison (1956). Laiho (1965), Chilvers and Pryor (1965) and Rambelli (1973) observed that fungal growth around and along roots precedes the infection, i.e. hyphal envelope must form before the intercellular penetration commences. This is confirmed in our work. The early view of Melin (1923) that intracellular infection would precede the formation of the Hartig net has already been refuted by Robertson (1954). Clowes (1951) claimed that mantle formation in *Fagus* preceded Hartig net formation. His conclusions were not based on continuous studies of the mycorrhizal formation process, however. The present work suggest that the labyrinthic inner mantle, when present, can be seen as an extension of the Hartig net. In cases where the mantle labyrinth tissue is very thick, or very thin or absent, this may be expressions of variable fungal sensitivity to the hypothetic morphogenic principle, or various levels of host production of it. The change in fungal morphology is not a prerequisite of penetration, but is a final result of it.

Clowes (1951) also found that in beech the rhizodermis had to be modified before fungal penetration could take place. Our study shows that in spruce, no such adaptation was necessary. On the contrary, the fungus was perfectly capable of invading all the mature cortex, but could not colonize the apical zone of differentiating cells, neither from outside nor from the infected cortex. Only later, when the mycorrhizal state was firmly established, did the root respond with retarded growth, markedly flourishing cells and increased vitality of the cortex

Table 1. *An interpretation of the phases of mycorrhiza formation in short roots*

Phase	Host action	Fungal action	Result
(1) Remote and selective stimulation of the fugus	Release of stimulatory compounds	Envigoured fungal growth, particularly in slow growing fungi (Chemotropism)	Contact between host and fungus
(2) Formation of a hyphal envelope	Release of stimulatory compounds (same as above?)	Aggregation of hyphae around the root	Formation of a fungal structure essential to infection
(3) Early penetration	Control of fungal lytic enzymes. No visible defence reactions	Withholding of disturbing enzyme activity. Mechanical penetration between cells	Fungal influence on host metabolism
(4) Labyrinthic Hartig net formation	The presence of the fungus induces release of factors affecting fungal morphogenesis	Morphogenetic change, formation into labyrinthic tissue	Formation of a new, 'symbiotic' organ: the mycorrhiza; its structure and mutual physiology is adapted to an efficient interchange of nutrients between host and fungus
(5) Mantle formation	Diffusion or translocation of factors affecting fungal morphogenesis	Labyrinthic growth is induced in all hyphae in physiological contact with the host	Host tissue slowly sealed off from soil contact by the growing mantle
(6) Completion of the mycorrhiza formation	Response to the presence of the fungus: retarded growth 'flourishing' of the cortex; dichotomy (in pine only)	Colonization of all mature cortical tissue (limitation by structural and physiological barriers; Nylund, unpublished)	Balanced relationship, mutual accommodation. All transport between host and soil passing through fungal tissue
(7) Dissolution of the mycorrhizal organ	Senescence and death of host cortex. Ended carbohydrate supply and control of fungal enzymes	Transition to saprophytic behaviour with intracellular penetration. Labyrinthic growth still maintained but less controlled	In the field: dissolution of the tissue by soil saprophytes. The stele remains living. *In vitro*: saprophytic colonization by the fungal symbiont which slowly dies. The distal end of the root continues mycorrhizal, while the proximal loses its cortex layer

(Hartley, 1969). This latter phenomenon has often been called hypertrophy but should preferably, in our opinion, be characterized as a lack of senescence and collapse of cortical cells, which is otherwise very common also in axenic, uninfected short roots.

The absence of any counteraction to the fungal invasion is interesting. Scannerini (1968) reported a strong 'hypersensitive' reaction in his naturally infected *Pinus strobus* roots. In fact the whole cortex of his mycorrhiza was dead, and heavily tannified. In the present study no sign of tannification, in addition to that with axenic roots, was detected in the healthy cortex; nor was any other histological change. This also being the rule in other studies examined, the ectomycorrhizal infection must be singled out as unique among fungal infections, not being counteracted by any of the ways, well known from plant pathology (Doak, 1955; Wheeler, 1975): tannification, wall thickening or papilla formation, necrosis, or disorganization and killing of fungal cells. The control mechanisms, indispensable in any kind of symbiosis, appear to be much more refined; namely control of fungal morphology, and suppression of lytic enzyme activity.

The mycorrhiza formation process appears to be controlled mainly by the host (Table 1), the host attracting the fungus, suppressing its lytic enzymes, inducing a morphological change, and setting limits to its growth. Of course, there will be continuous interactions between the organisms, which may account for the variation in mycorrhizal structure in the same host with different mycobionts, but these are secondary to the overall control exercised by the host. The control measures may be pre-determined and not necessarily induced by the presence of the fungus: in a phylogenetic perspective, however, they are meaningful and hardly incidental, as we must assume a very long co-evolution between vascular plants and a sequence of mycobionts (Nicholson, 1975).

Firstly, the roots stimulate the tentative mycobionts by some generally produced metabolite, the 'M-factor', to which this group of fungi have developed responses: increased growth, a character variable with the species (Melin, 1963), and the formation of a hyphal envelope, a character common to all (Marks and Foster, 1973). The mycorrhizal fungi recognize only the signal of a root in general, not the species of host. The mycorrhizal fungi alone among all members of the rhizosphere flora have 'learned' the recognition process.

The next factor regulating the mycorrhizal infection is the 'general defence barrier', necessarily upheld by any root in a soil full of tentative decomposers. This barrier is demonstrated by the inability of the parasite *H. annosum* and the potent saprophytes, *T. viride*, *M. radicis atrovirens* and the *Penicillia* to infect roots, as demonstrated in the present work. As the root apical meristem and cortex lack cuticle and bark layer, structures protecting other plant parts, the barrier must have the character of chemical action rather than of a structural obstacle; again, the mycorrhizal fungi have adapted to overcome it. Their broad host range, and the compatability between host and fungal species from different areas indicate, as in the case of the M-factor, that the barrier must be similar in all host plants. Again, no single species of host has to be 'recognized' as such.

Thirdly, the production of fungal enzymes capable of lysing host cell walls is not triggered, although the substrate is present, and the fungus does not 'disturb' the host with lytic enzyme activity as do many plant pathogens (Wheeler, 1975)*.

* In spite of the failure to demonstrate such enzymes *in vitro* (Lindeberg and Lindeberg, 1977) not only the above cited observations, but also the work of Filer and Toole (1966) prove that ectomycorrhizal fungi have a capacity to perforate cell walls under certain conditions.

The lack of visible host reaction to the fungal penetration indicates that the host does not immediately 'notice' the presence of the fungus. The lytic enzyme suppression mechanism may again be due to pre-formed factors, the fungus being adapted to regulate its enzyme production as a response to them.

The last phase of the infection process, the morphogenesis into labyrinthic tissue, is accomplished only slowly, and does not commence on the root surface but within the cortex. This indicates some kind of continuous metabolic reaction by the host, requiring a period of induction in the presence of the fungus. Here, we may have the first induced response by the host to the fungal infection. Even in this case, there must be an influence on the fungus of a very general character: different coniferous and angiosperm hosts induced the same type of morphogenesis in the fungus and, more remarkably, the same morphogenetic response in a number of very different fungal taxa. Further research on this phenomenon may provide a key to a deeper understanding of fungal morphogenesis.

During further development of the mycorrhiza many more mutual interactions will take place. However, during this first period of contact and infection in a previously uninfected host, the following simple relations seem to determine the process: The host is phylogenetically adapted to the state of ectomycotrophy, and built-in controls regulate the fungal infection; the presence of the fungus only slowly leads to a response, and the fungus is never 'recognized' specifically. The fungus is also adapted to a symbiotic existence. This includes adaptation to a set of conditions common to all host species in its range, but again, no specific recognition process in the fungus is required as it is in for example in many plant diseases or in legume infections by *Rhizobia* (Heslop-Harrison, 1978; Vanderplank, 1978, Bauer, 1981).

This proposed model, although rather speculative, not only accommodates the observations made in the present study, but would also account for the great intercompatibility between ectomycorrhizal hosts and fungi (Trappe, 1962). The exceptions, on the other hand, where the fungus seems to be strongly host specific, may in some cases be a result of ecological rather than physiological specificity (possibly in the well known pair *Suillus grevillei* – Larch), or be a result of further specialization (e.g. in *Alnus*, cf. Molina, 1981).

In a greater perspective, the host must be regarded as controlling in rather great detail the character of a mycorrhizal relationship, the fungus only being able to express its specific character within the set frames. *P. croceum* forms a typical coniferous ectomycorrhiza with pine and spruce. In birch and beech (Nylund, unpublished), it only colonized the radially elongated rhizodermis intercellularly, but in *Arbutus* (Zak, 1976) it formed typical arbutoid ectendomycorrhiza, the same strain being compatible with spruce (Nylund, 1981). Similar observations were made concerning *Amanita* gemmata (Largent *et al.*, 1980) which was ectomycorrhizal on *Pinus contorta* but ectendomycorrhizal (*Arbutus* type) on *Arctostaphylos*. In a remarkable but little noticed study, Filer and Toole (1966) made a number of well-known ectomycorrhiza formers to produce endomycorrhiza with the normally VA-mycorrhizal *Liquidambar styraciflua*. Evidently, ectomycorrhizal fungi are able to adapt in several different ways ontogenetically, at least *in vitro*. These modes of adaptation are independent of ecological factors; the type of mycorrhiza being genetically determined and controlled by the host, partly through structural dispositions (Nylund, in prep.), and partly by active physiological reactions to the fungal infection.

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